

Victor Chua

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Singaporean Citizen | [linkedin.com/in/vicchua123](https://www.linkedin.com/in/vicchua123) | Summer 2027 Internship: May 11 – August 11, 2027

To: Selection Committee, Engineering Competence Centre (ECC) R&D

Company: Thales Singapore

Date: July 22, 2026

Subject: Application for Software Engineer Intern (AI)

Dear Members of the Selection Committee,

I am writing to express my strong interest in the **Software Engineer Intern (AI)** position at Thales Singapore's Engineering Competence Centre (ECC). As a Computer Science undergraduate at Nanyang Technological University in the Turing AI Scholars Programme (TAISP), my research and engineering focus centers on industrializing Retrieval-Augmented Generation (RAG) and Agentic AI architectures. Thales's mission of building trustworthy, human-centric technology directly aligns with my goal of engineering high-reliability AI systems for critical applications.

During my Machine Learning Internship at Yap Motor, I was tasked with building the company's first AI system from scratch. I architected and deployed a production-ready RAG system leveraging Google Gemini and ChromaDB vector embeddings, serving 24 automotive mechanics daily across 8 diagnostic categories. To handle complex query edge cases, I engineered custom vector search re-ranking with regex NLP metadata extraction and semantic document chunking. This experience taught me how to take open-source AI technology stacks from initial prototype to dependable industrial deployment.

Complementing my RAG deployment background, I have developed multimodal AI pipelines and agentic architectures across research and product environments. At SUSS Digital Solutions & AI, I engineered an on-premises LLM inference pipeline for privacy-sensitive university documents and integrated TwelveLabs multimodal video analysis for automated content categorization. Additionally, for my personal project (Drum Hub), I designed an Agentic AI system using LangGraph with explicit state machines and tool-calling schemas, while training a dual-head MobileNetV3 vision model published on HuggingFace.

I am eager to contribute my hands-on RAG, Agentic AI, and Python pipeline engineering skills to the Thales ECC R&D team for the full 2027 summer internship period (May 11 – August 11, 2027). Thank you for your time and consideration.

Sincerely,

Victor Chua

Student, Turing AI Scholars Programme (TAISP)

Nanyang Technological University (NTU)